P-D-223C
August 16, 1978
SUPERSEDING
Fed. Spec. P-D-223B
May 9, 1975

FEDERAL SPECIFICATION

DETERGENT, GENERAL PURPOSE, LIQUID, HEAVY-DUTY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers commercial-type liquid detergents for heavy-duty general purpose cleaning (see 6.1).

1.2 Classification.

1.2.1 Classes. The general purpose detergents shall be of the following classes as specified (see 6.2).

Class 1 - Heavy-duty, built, cleaner.
Class 2 - Heavy-duty, solvent, cleaner.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-C-96 - Cans, Metal, 28-Gage and Lighter.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-P-704 - Pails, Metal (Shipping, Steel, 1 through 12 Gallons).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 7930

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(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Appearance. The classes 1 and 2 products shall be stable, homogeneous liquids and free of sediments.

3.2 Foam height. The foam height of the liquid cleaners in distilled water shall be not greater than 5.0 centimeter (10 seconds value), when tested as specified in 4.3.5.

3.3 pH value. The pH of a 1.0 percent by volume solution of the product in distilled water shall be not less than 9 nor more than 12, when tested as specified in 4.3.1.

3.4 Rinsing properties. The product shall be free-rinsing, when tested as specified in 4.3.2.

3.5 Cleaning efficiency. A solution of one-half percent by volume (one-half volume of the product to 99.5 volumes of hard water) shall exhibit a cleaning efficiency of not less than 90 percent, when tested as specified in 4.3.3.

3.6 Non-volatile matter. The classes 1 and 2 products shall have a non-volatile content of not less than 17.0 and 20.0 percent respectively, when tested as specified in 4.3.4.

3.7 Deleterious action on painted surfaces. The difference between the percent specular loss of a 0.2 percent solution of trisodium phosphate minus the percent specualr loss of a 0.2 percent solution of the product shall not exceed 0.5 the percent specular loss of the trisodium phoshate, when tested as specified in 4.3.6.

3.8 Deleterious action on linoleum, asphalt, rubber, and vinyl-asbestos tile. A 2.0 percent by volume solution of the product shall not cause any softening, swelling, cracking, or discoloration of the test specimens, when tested as specified in 4.3.7.

3.9 Flash point (class 2). The flash point of the product shall not be less than 79°C (175°F), when tested as specified in 4.3.8.

3.10 Use labels. The manufacturer's recommended dilutions and "Direction For Use" shall be clearly shown on the label of each container.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Quality conformance inspection. Sampling for inspection and acceptance shall be performed in accordance with MIL-STD-105. The end item shall be examined in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL) set forth below. The sample unit shall be one shipping container fully prepared for delivery.

4.2.1 Examination of the preparation for delivery. Packing, and marking shall comply with the requirements of section 5. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.2.2 Testing the end item. The end item shall be tested for compliance with the requirements of section 3. The lot shall be considered unacceptable if sample fails any test. Sampling shall be in accordance with inspection level S-2.

4.2.3 Examination of the end item container for defects in appearance of product. The inspection level shall be S-2 with an AQL or 2.5 percent defective. An examination shall be made to determine with requirements under 3.1.

4.3 Test methods.

4.3.1 pH value. Prepare a 1.0 percent solution of the product in freshly boiled distilled water which has been cooled to 23°C ± 3°C (73.4°F ± 5°F). Determine the pH of this solution by means of a pH meter which has been previously standardized with standard buffer solutions at pH 10.0 and 7.0. No sodium ion concentration correction is necessary.

4.3.2 Rinsing properties. Make a solution of the product at the most concentrated manufacturer's recommended dilution in enough synthetic hard water (see 4.3.2.1) to give 100 ml of total solution. Pour the resultant solution into a clean, dry 250-ml Erlenmeyer flask. Stopper the flask and shake vigorously for 1 minute. Pour out the solution and rinse the flask three times with 75-ml portions of synthetic hard water. Invert the flask and allow to dry. Examine for any residue. The flask shall contain no more residue than a flask allowed to dry after rinsing with synthetic hard water alone.

4.3.2.1 Synthetic hard water (300 ppm hardness). To a 1-liter volumetric flask, add 0.246 g calcium chloride dihydrate ($CaCl_2.2H_2O$) and 0.295 g magnesium sulfate heptahydrate ($MgSO_4.7H_2O$). Add a small portion of distilled or deionized water to dissolve the salts; then dilute the solution to the mark with distilled water.

4.3.3 Cleaning efficiency.

4.3.3.1 Testing procedure. This procedure shall be carried out five times for each cleaning solution tested, using five separate test panels.

4.3.3.1.1 Panel precleaning. Cut an 18 inch piece from the supply of coverbase to fit the washing tray of the Gardner Washability machine. The 18 inch panel shall then be scrubbed by hand using a cellulose sponge saturated with a solution consisting of a 0.5 wt percent solution of linear alkylaryl sodium sulfonate (85 percent active) in tap water. Rinse the panel under cool running tap water and then drain for 10 minutes. Blot the remaining water with a paper towel. Oven dry for 15 minutes at 100°C and then allow to cool at room-temperature. Determine the reflectance of the panel (see 4.3.3.2.1) and record.

4.3.3.1.2 Panel soiling.

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4.3.3.1.2.1 Preparation of soiling mixture.

Soiling mixture (see 3)	Parts by weight
Metallic Brown	20
Odorless Kerosene	12
Stoddard Solvent	12
White Mineral Oil (USP)	1
Lubricating Oil (SAE 10)	1
Hydrogenated Vegetable Shortening	1

Combine oils and shortening and heat to 41°C ± 1°C to melt. Add half the kerosene. With a spatula, mix in the metallic brown. Add remaining kerosene and Stoddard solvent, and mix for 2 hours with a magnetic stirrer.

4.3.3.1.2.2 Soil procedure. While the bulk lot of soil is continuously stirred to insure uniformity, a sample of 5 g is removed and applied along entire length of the vinyl panel. A Byrd applicator or "Doctor" blade, set at 0.008 inch clearance and 2 inches width, is used to provide an even spread.

After soil application, air-dry panels at least 30 but not more than 60 minutes, then cure panels on smooth surface in a forced draft oven at 100°C ± 3°C for 60 minutes. Remove the panels from oven and cool to room-temperature, using the panels within 24 hours.

4.3.3.1.3 Panel washing.

4.3.3.1.3.1 Preparation of test detergent. Synthetic hard water shall be prepared according to the following formula:

0.132 gram $CaCl_2.2H_2O$ (CP).

0.1475 gram $MgSO_4.7H_2O$ (CP).

The hardwater salts shall be dissolved in a small portion of distilled water, the solution is transferred quantitatively to a liter volumetric flask and the flask is filled to volume. Liquid products shall be tested in hard water at the recommended use concentration.

4.3.3.1.3.2 Washing procedure. The sponge, when dry, shall be cut to size to fit the sponge box. The thickness of the sponge shall be 1 inch when wet. Total weight of the box and dry sponge shall be 1 pound ± 1/2 ounce. Soak panel for 60 seconds in wash solution sufficient to cover entire panel. Wet sponge with 25°C ± 3°C tap water and sqeeze damp-dry. Add 50 ml of detergent solution to sponge and insert sponge in box. Center test panel in washability pan with two properly sized pieces of vinyl molding. Start apparatus set at one stroke per second (one cycle back and forth is two strokes), while dripping detergent solution from a pipet onto test panel center at a rate of 12 ml of solution during the wash period of 100 strokes. Stop apparatus, remove and rinse test panel under a light stream of 25°C ± 3°C tap water. Drain and replace panel in apparatus in the reverse direction. Rinse sponge in 25°C ± 3°C tap water, squeeze damp-dry, add 50 ml of detergent solution to unused side, replace sponge in box and repeat 100 stroke wash cycle as before. Rinse panel and air dry. Determine the reflectance of the washed portion of the panel and record.

4.3.3.2 Evaluation procedure.

4.3.3.2.1 Reflectance measurement. Using a photometer and search unit equipped with tristimulus green filter, reflectance of the surface of the test panel is measured before soiling and after washing using a vinyl panel template of the dimensions given in figure I. Four reflectance measurements shall be made of different areas of the soiled and washed portion of each panel examined. The four readings are measured to the nearest 0.5 percent, and averaged. This procedure shall be repeated for each of the five panels required to test each cleaner. Between panels, the search unit is rested on a standard white plaque of approximately 80 percent reflectance and the instrument checked before each series of readings.

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4.3.3.2.2 Cleaning efficiency (C.E.)

$$\text{Percent cleaning efficiency} = \frac{R_2}{R_1} \times 100$$

Where R_1 = average reflectance of unsoiled, precleaned panel.
R_2 = average reflectance of soiled, test washed panel.

Report the cleaning efficiency results as the average of the cleaning efficiencies obtained from the five test panels. The results shall be reported to the nearest unit percent.

4.3.3.3 Special apparatus and materials.

4.3.3.3.1 Reagents.

Linear dodecylbenzene sodium sulfonate (85 percent active)

Supplier: Witco Chemical Corporation

Metallic Brown

Supplier: General Services Administration
Federal Supply Service
Chemical Branch (FMBC)
Washington, D.C 20406

Hydrogenated Vegetable Shortening

Kerosene, Odorless

Lubricating Oil (SAE 10)

Sodium Sulfate, Anhydrous

Stoddard Solvent

White Mineral Oil (USP)

4.3.3.3.2 Straightline Washability Apparatus (Henry A. Gardner Laboratory, Inc., Bethesda, MD 20014).

4.3.3.3.3 Photovolt Model 670 or equal with a 610Y search unit and tristimulus green filter (range 520 to 540 millimicrons) manufactured by Phtovolt Corp., 1115 Broadway, New York, NY 10010 has been found suitable.

4.3.3.3.4 Gardner adjustable blade ("doctor" blade) Model AG 3820 A (Henry A. Gardner Laboratory, Inc., Bethesda, MD 20014).

A substitute for the "doctor" blade is the Byrd Applicator AR-3800 of 2 inch width and a clearance of 8 mils (Henry A. Gardner Laboratory, Inc., Bethesda, MD 20014).

4.3.3.3.5 White Vinyl Cove Base.

General Services Administration
Federal Supply Service
Chemical Branch (FMBC)
Washington, D.C. 20406

4.3.3.3.6 Forced draft oven having + 2°C tolerance.

4.3.3.3.7 Cellulose sponge, photographic grade (fine pore).

4.3.3.3.8 Standard white plaque (reflectance approximately 80 percent).

4.3.3.3.9 Template of vinyl panel, 2 inches by 4 inches with four evenly spaced holes for making reflectance readings. See figure I.

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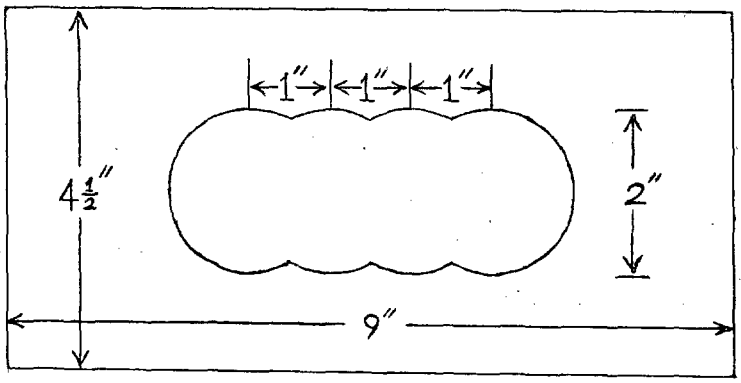


Figure I. Template Dimensions

4.3.4 Non-volatile matter. Weigh an aluminum pan (12.7 cm. in diameter by 3.2 cm in depth) to the nearest mg. Weigh 2.0 g of the sample to the nearest mg into the tared pan. Place pan and sample in an air oven at a temperature of 105°C ± 2°C (221°F ± 4°F). After heating for exactly 4 hours, cool in a desiccator and weigh to the nearest mg. Calculate the percentage of non-volatile matter. After heating for 4 hours, Class 2 sample leaves a vicous non-volatile liquid.

4.3.5 Foam height (class 1). Dilute the sample 1 to 30 with distilled water. Place 100 ml of the solution at 54.4°C (130°F) in a 100-ml glass stoppered graduated cylinder. Shake vigorously for 5 seconds and place at rest for 10 seconds. Measure the foam height.

4.3.6 Deleterious action on painted surfaces.

4.3.6.1 Measurement of specular gloss. Gloss shall be determined using a glossmeter. A list of manufacturers of glossmeters can be obtained from ASTM Headquarters, 1916 Race Street, Philadelphia, PA 19103. Operate the glossmeter in accordance with the manufacturer's instruction.

4.3.6.2 Test panels. Stainless steel, aluminum, or glass panels, approximately 3 inches by 6 inches by 1/16 inch (or thinner) shall be used. The panel shall be prepared by applying (brush or spray) one coat of each of the following paints: (1) enamel, undercoat, white as the first coat; (2) enamel, semi-gloss, white with a 60° gloss of 40 to 60 as the final coat. A drying time of not less than 24 hours between coats and a minimum of not less than 24 hours after the application of the final coat shall be allowed. When tests are to be made on panels that have been used, roughen the surface (to obtained good adhesion) before applying the final coat (60° gloss, 40 to 60, white). Allow the final coat to dry for 1 day minimum or 4 days maximum.

4.3.6.3 Procedure. Prepare one liter each of a solution of trisodium phosphate (sodium phosphate, tribasic anhydrous) and a of the cleaning compound under test, at concentrations of 2 g per liter in distilled water. Measure the specular gloss of two panels and then immerse one panel in each of these solutions, kept at room temperature, for 1 hour. Remove the panels, rinse thoroughly in distilled water, and dry at room temperature for at least 1 hour. Again measure the specular gloss of each panel. The percentage decrease in gloss of the portion of the panel immersed, compared with the gloss of same panel before immersion, shall be reported for the compound under test and for the trisodium phosphate.

4.3.7 Deleterious action on linoleum, asphalt, rubber, and vinyl-asbestos tile flooring.

4.3.7.1 Test panels. These panels shall be as specified below:

White linoleum panels - 2 by 4 by 1/8 inches.
Black asphalt panels - 2 by 4 by 1/8 inches.
Black rubber tile panels - 2 by 4 by 1/8 inches.
Black vinyl-asbestos tile panels - 2 by 4 by 1/8 inches.

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4.3.7.2 Procedure. Pour 150 ml of a 2.0 percent (by volume) of the cleaning compound solution into each of the four 250-ml beakers. Immerse one panel of each in the beakers containing the cleaning solution. Allow the panels to remain in the solution at room temperature (23°C ± 3°C) for 16 hours. At the end of this time, remove the panels, rinse thoroughly with distilled water, and dry at room temperature for at least 1 hour. Examine for softening, swelling, cracking, bleaching, and discoloration.

4.3.8 Flash point (class 2). The flash point shall be determined using a Tag Open-Cup Flash Tester. A list of manufacturers of Tag Open-Cup Flash Tester can be obtained from ASTM Headquarters, 1916 Race Street, Philadelphia, PA 19103. Operate the apparatus in accordance with the manufacturer's instruction.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One-gallon quantity. One gallon of detergent of one class only shall be packaged in a can conforming to PPP-C-96, type V, class 4. The can shall be closed in accordance with the appendix of the can specification.

5.1.1.2 Five-gallon quantity. Five gallons of detergent of one class only shall be packed in a pail conforming to PPP-P-704, type 1, class 8. The pail shall be closed in accordance with the appendix of PPP-P-704.

5.1.1.3 Class 1. Fifty-five gallons of detergent of one class only shall be packaged in a drum conforming to PPP-D-729, type I or II.

5.1.2 Commercial. The detergent shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One-gallon quantity. Six one-gallon containers of one class only shall be packed in accordance with PPP-C-96 for level A packing requirements.

5.2.1.2 Five-gallon quantity. Five gallon containers do not require packing.

5.2.1.3 Fifty-five quantity. Fifty-five gallon containers do not require packing.

5.2.2 Commercial. The detergent shall be packed in containers to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended uses.

6.1.1 Class 1. The concentrated liquid detergent covered by this specification is intended for use as a general purpose cleaner. The product is suitable for most cleaning operations and should be diluted according to instructions on label. It can be used to clean floors, walls, woodwork, painted surfaces, concrete, tile, porcelain, glass, fixtures, vinyl plastic upholstery, etc., and can be used in automatic scrubbing machines.

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6.1.2 Class 2. The solvent-type liquid detergent covered by this specification is intended for heavy duty cleaning operations (e.g., floors, walls, concrete, general maintenance, etc.) where grease, oil, sludge, and carbonaceous deposits are encountered.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity required.
(c) Packaging and packing required (see 5.1 and 5.2).
(d) Class required (see 1.2).
(e) Marking required (see 5.3).

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

COM - NBS
GSA - FSS
NASA - MSF
VA - DMS

User activity:

Army - GL

PREPARING ACTIVITY: GSA - FSS

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